

KRISTIYAN SAKALYAN

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[Page](#) ◊ [Scholar](#) ◊ [Github](#) ◊ [LinkedIn](#)

EDUCATION

Technical University of Munich (TUM)

September 2025 (*expected*)

M.Sc. in Data Engineering & Analytics

Expected average grade: 1.2

Current rank: Top 4%

Technical University of Munich (TUM)

October 2022

B.Sc. in Computer Science

Thesis title: "Multi-object tracking for annotating markers for a motion capture system"

Thesis grade: 1.3

RESEARCH INTERESTS

- AI4Science
- Generative AI
- Graphs and Geometric Deep Learning

PUBLICATIONS

- [1] **K. Sakalyan***, A. Palma*, F. Guerranti*, F.J. Theis, S. Günnemann. "Modeling Microenvironment Trajectories on Spatial Transcriptomics with NicheFlow." In ICML GenBio Workshop, 2025.

RESEARCH EXPERIENCE

Modeling Microenvironment Trajectories on Spatial Transcriptomics Jan 2024 – Present
Supervisors: [F. Guerranti](#), [A. Palma](#), [F. Theis](#), [S. Günnemann](#) [TUM](#)

- Investigating graph- and point cloud-based methodologies to model single-cell dynamics, leveraging Flow Matching and Optimal Transport for time-resolved spatial data.

Conditional Flow Matching for Temporal Graphs Oct 2024 – Jan 2024
Supervisors: [F. Guerranti](#), [A. Palma](#), [S. Günnemann](#) [TUM](#)

- Extend previous works by applying flow matching to dynamic temporal graphs. [[Code](#)]

Continual Learning for Domain Incremental Semantic Segmentation Apr 2024 – Oct 2024
Supervisors: [G. Ghazaei](#), [R. Avecado](#), [M. Fornasier](#) [ZEISS](#) & [MDSI](#) & [TUM](#)

- Developed a new method to reduce catastrophic forgetting in domain-incremental semantic segmentation using contrastive learning [[Report](#)]

Explainable Document Embeddings Oct 2023 – Apr 2024
Supervisors: [G. Hagerer](#), [G. Groh](#) [TUM](#)

- Enhanced the Bag-of-Concepts framework with an automated approach to concept labeling for document embeddings using LLMs [[Code](#)]

Patch by Patch Coarse-to-Fine 3D Shape Refinement

Oct 2023 – Feb 2024

Supervisors: [L. Li](#), [M. Nießner](#)

[TUM](#)

- Developed a method to enhance geometric detail in 3D shape generation through patch-based coarse-to-fine refinement of sparse point clouds [[Report](#), [Poster](#)]

Multi-object tracking for annotating markers for a motion capture system

Nov 2021 – Mar 2022

Supervisors: [S. Banik](#), [A.C. Knoll](#)

[TUM](#) & [reFit](#)

- Conducted research on automating marker annotation for custom skeletons, integrating AI and motion capture systems as part of my Bachelor's thesis at TUM and reFit Systems GmbH.

ACHIEVEMENTS

Deutschlandstipendium

Oct 2023

Awarded by TUM and Infineon Technologies AG for outstanding academic achievements. [[PDF](#)]

WORK EXPERIENCE

Machine Learning Engineer

Jan 2025 - Present

Working Student

[Rohde & Schwarz](#)

- Working on the computer vision method used in [QPS Walk 2000](#).

Data Engineer & Scientist

Mar 2023 - Jan 2025

Working Student

[Airbus Defence & Space](#)

- Conducted machine learning research on 4D flight trajectory prediction, implementing transformer-based, LSTM, Neural ODE, LSTM-ODE hybrid, Conditional Flow Matching, and Diffusion models.
- Engineered data pipelines with Dask for large-scale processing and redesigned real-time prediction services for airplane ETA using Python, Docker, Kubernetes, Helm, TerraForm, Azure Pipelines, Grafana, Prometheus.

Software Engineer

Aug 2022 - Feb 2023

Working Student

[Techpilot](#)

- Led sprint planning, code reviews, and iterative delivery in a collaborative agile environment.
- Enhanced the web platform by building features with React, TypeScript, and Material-UI, connecting buyers with suppliers.

Junior Software Engineer

Oct 2020 - Jul 2022

Working Student

[Techpilot](#)

- Maintained and expanded client-server applications, applying React, TypeScript, and Spring Framework for scalable development.
- Contributed to front-end and back-end enhancements, optimizing user experience and system efficiency.

R&D Engineer

Dec 2019 - May 2020

Working Student

[HARMAN International](#)

- Developed a POI aggregation web service prototype, consolidating and visualizing location data from various sources on a map.
- Designed algorithms to eliminate duplicates, ensuring comprehensive, non-redundant data aggregation.

SKILLS/HOBBIES

Programming Languages	Python, C++, SQL, Java, JavaScript, TypeScript
Machine Learning Tools	Pytorch, Sklearn, Pandas, Polars, Numpy, TensorBoard, W&B
	Hydra, Slurm
	PySpark, Dask
Cloud & Infrastructure	Azure, Azure Pipelines, Terraform, Kubernetes, Docker, Helm
Monitoring Tools	Grafana (Loki, Tempo), Prometheus
Databases	PostgreSQL, SQLite, MongoDB, Redis, RisingWave
Web	React.js, Material-UI, Spring, Express, Node.JS
Hobbies	hiking, running, cycling, swimming, handcrafting

EXTRACURRICULAR TRAINING

Natural Language Processing Specialization	Oct 2023
· Certificate	
Geometric Deep Learning	Oct 2024
· Lecture and Exercises	

LANGUAGES

Bulgarian	Native	German	C1
English	Proficient	Spanish	Beginner